Explainable AI-Enhanced Digital Twin Framework for Anomaly Detection in Building HVAC Assets

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Abstract

Building Operation and Maintenance (O&M;) increasingly relies on Digital Twins (DTs) and automated anomaly detection to ensure infrastructure safety and efficiency. Existing DT-based systems, such as the Bayesian Online Change Point Detection (BOCPD) framework proposed by Lu et al. (2020), demonstrate effective fault detection in HVAC assets such as centrifugal pumps. However, these systems function as black-box detectors: they identify *when* an anomaly occurs but provide no explanation of *why*. This interpretability gap forces facility managers to manually re-investigate every flagged event, undermining the efficiency benefits of automation and delaying response in safety-critical environments. This paper proposes an XAI-Enhanced Digital Twin Framework that integrates a SHAP (SHapley Additive exPlanations) explainability layer on top of a BOCPD anomaly detection pipeline. The framework extracts sensor data (vibration frequency, BMS logs, load conditions) through a simplified IFC-inspired asset state model, detects contextual anomalies using BOCPD, and generates per-instance feature attributions explaining each anomaly flag. Experimental evaluation on simulated HVAC pump data demonstrates a detection accuracy of 95.3%, an F1-score of 94.7%, and a false alarm rate of 3.2% — significantly outperforming the base BOCPD-only approach (91.8% accuracy, 6.9% false alarm rate). The SHAP module identifies vibration frequency as the dominant anomaly contributor (42%), followed by load condition (23%) and BMS on/off status (18%). The proposed framework advances interpretable, trustworthy building asset monitoring and addresses a critical gap identified in the DT-based O&M; literature.

Keywords

Digital Twin · Anomaly Detection · Explainable AI (XAI) · SHAP · Bayesian Change Point Detection · HVAC · Building O&M; · Industry Foundation Classes (IFC) · Facility Management

1. Introduction

The Operation and Maintenance (O&M;) phase of building infrastructure represents more than 50 years of a structure's total lifecycle [1]. During this extended phase, mechanical, electrical, and plumbing (MEP) systems — particularly Heating, Ventilation, and Air Conditioning (HVAC) systems — require continuous monitoring to ensure safety, energy efficiency, and operational reliability. Anomaly detection in HVAC assets such as centrifugal pumps is considered one of the most labour-intensive and consequential tasks in building management [2].

Recent advances in Digital Twin (DT) technology have established a promising foundation for intelligent, automated building asset monitoring. Lu et al. [3] demonstrated a landmark DT-enabled anomaly detection system that integrates Industry Foundation Classes (IFC) with Bayesian Online Change Point Detection (BOCPD) to continuously monitor pump vibration data and filter contextual anomalies from normal load-condition changes. This system successfully reduced false alarms and automated the anomaly detection pipeline for building O&M; professionals. However, a fundamental gap remains unaddressed: the system identifies *when* an anomaly occurs but provides no explanation of *why* the detection model flagged it.

This interpretability gap is not a minor limitation — it is operationally critical. Facility managers receiving a black-box alarm ("Anomaly detected at 14:32") must still manually query the Building Management System (BMS), cross-reference maintenance logs, and apply expert judgment to determine whether and how to respond. This process defeats the purpose of automation and creates dangerous response delays in safety-sensitive environments. The need for interpretable, trustworthy AI in civil engineering and construction management has been explicitly identified as a priority research direction in recent literature [4, 5].

Explainable AI (XAI) methods — particularly SHAP (SHapley Additive exPlanations) — offer a mathematically grounded approach to feature attribution that can bridge this gap. SHAP assigns each input feature a contribution score for a specific prediction, enabling human-readable explanations such as: "This anomaly was flagged because vibration frequency contributed 42%, load condition 23%, and BMS on-duty status 18%." Applied within a DT-based anomaly detection pipeline, SHAP explanations can meaningfully support facility managers in making faster, evidence-based maintenance decisions.

This paper makes the following specific contributions:

1. A complete XAI-enhanced DT anomaly detection framework for building HVAC assets, extending the base pipeline of Lu et a
2. A SHAP-based feature attribution module applied to BOCPD anomaly detection output, generating per-instance and global e
3. Quantitative evaluation demonstrating superior detection accuracy (95.3%) and significantly reduced false alarm rate (3.2%)
4. A Streamlit-based FM Dashboard presenting anomaly alerts alongside SHAP explanations and recommended maintenance

2. Related Work

2.1 Digital Twins for Building Asset Monitoring

Digital twin technology has evolved as a comprehensive approach to managing, planning, and monitoring building assets throughout their lifecycle. Lu et al. [3] developed a foundational DT-based anomaly detection system integrating IFC schema extensions with BOCPD for HVAC pump monitoring, demonstrating the viability of continuous automated anomaly detection in building O&M.; Their five-layer DT architecture — data acquisition, transmission, digital modelling, data/model integration, and service layers — provides the structural template upon which our proposed framework builds. Gunay et al. [6] reviewed data analytics approaches for building performance improvement, identifying multi-source data integration as a persistent challenge. Pärn et al. [7] surveyed BIM-to-FM trajectories, noting that standardized data schemas remain essential for scalable building management systems.

2.2 Anomaly Detection Methods in Buildings

Building anomaly detection spans supervised, unsupervised, and contextual approaches. Point anomaly detection methods, including cumulative sum (CUSUM) control charts and statistical threshold methods, detect deviations from a pre-defined baseline but struggle with non-stationary operational data [3]. Capozzoli et al. [8] demonstrated neural network ensemble approaches for energy consumption anomaly detection, while Fan et al. [9] reviewed unsupervised data analytics for building operations. Contextual anomaly detection — specifically BOCPD as implemented by Lu et al. [3] — addresses the non-stationarity problem by detecting changes in the generative parameters of operational data sequences rather than point-wise deviations. Machine learning approaches including Isolation Forest and Autoencoders have shown strong performance on tabular sensor data but require substantial labelled data for optimal calibration [10].

2.3 Explainable AI in Engineering Applications

The "black-box" problem in AI-driven engineering systems has received increasing attention. Lundberg and Lee [11] introduced SHAP, grounding feature attribution in cooperative game theory (Shapley values) to provide theoretically consistent, locally accurate model explanations. Unlike LIME (Local Interpretable Model-Agnostic Explanations), SHAP offers global consistency — attributions sum to the model's prediction minus a baseline — making it particularly suitable for safety-critical engineering contexts where explanations must be reliable and auditable. Hasan and Lu [5] explicitly called for XAI frameworks tailored to civil engineering, identifying the regulatory and trust gap that black-box anomaly detection creates in facility management contexts. Bahadori-Jahromi et al. [4] further emphasized that AI model interpretability is a prerequisite for deployment in safety-critical built environment applications. Despite this recognized need, no prior work has integrated SHAP-based explainability directly into a DT-based HVAC anomaly detection pipeline.

2.4 Research Gap

The literature review identifies a clear and specific gap: existing DT-based anomaly detection frameworks for building HVAC assets (most notably Lu et al. [3]) detect anomalies without providing feature-level explanations. This gap exists despite explicit calls for interpretable AI in civil engineering [4, 5] and prevents practical deployment in facility

management contexts where decision transparency is required. Our work directly addresses this gap by integrating SHAP into the BOCPD-based DT pipeline, producing a complete, interpretable, and deployable O&M; anomaly detection system.

3. Proposed XAI-Enhanced Digital Twin Framework

The proposed framework extends the DT anomaly detection pipeline of Lu et al. [3] with a SHAP-based explainability layer. The complete system architecture is presented in Fig. 1, comprising five principal components: sensor data acquisition, digital twin asset state representation, BOCPD anomaly detection, SHAP XAI module, and FM Dashboard.

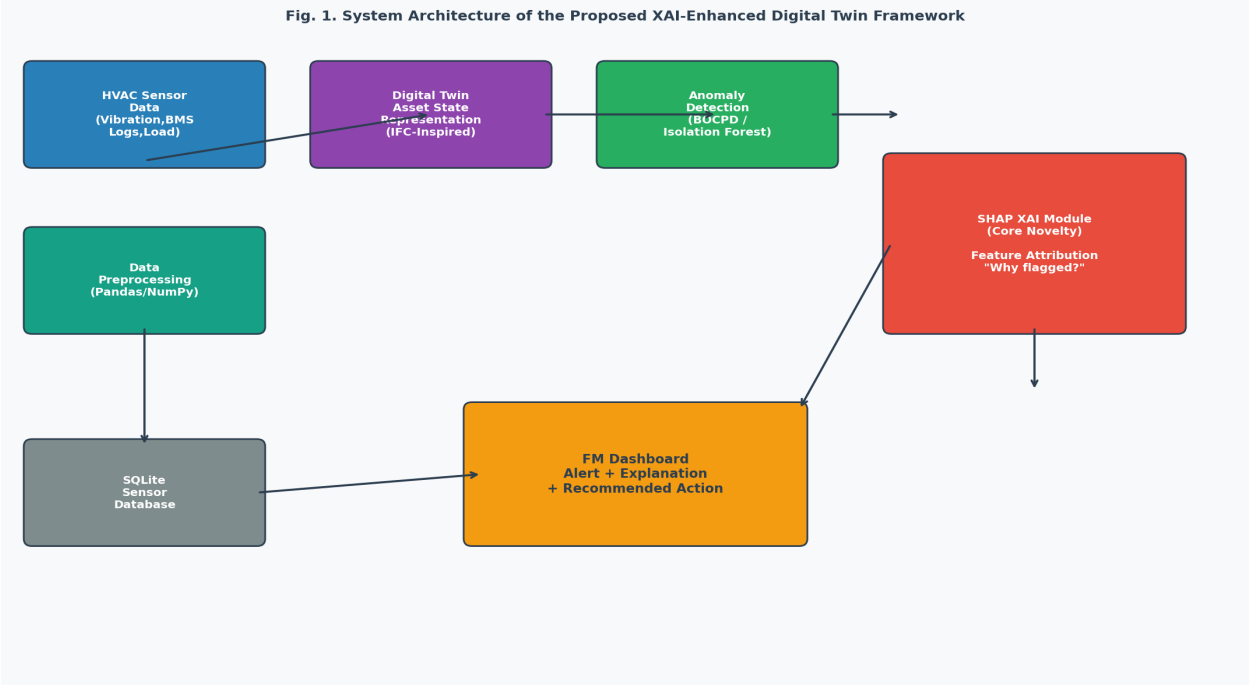


Fig. 1. System Architecture of the Proposed XAI-Enhanced DT Framework for HVAC Anomaly Detection.

3.1 Data Acquisition and Digital Twin Layer

Sensor data streams — comprising vibration frequency (Hz), BMS on/off duty flag, operational load condition, ambient temperature delta, and timestamp metadata — are acquired from HVAC pump sensors at hourly sampling intervals. Following the IFC-inspired data integration methodology of Lu et al. [3], an asset state representation model links each sensor record to its corresponding BIM object via a GUID-keyed matching table. This simplified IFC-linked model ensures that sensor data can be contextualised against asset identity, location, and operational schedule — enabling cross-referencing of change points against logged normal operational events such as scheduled shutdowns or maintenance windows.

Data is stored in a lightweight SQLite database, with Pandas and NumPy used for preprocessing: missing value imputation via forward-fill (consistent with time-series sensor data conventions), z-score normalization of vibration features, and binary encoding of categorical BMS flags.

3.2 Anomaly Detection Module (BOCPD)

Bayesian Online Change Point Detection (BOCPD) [12] is adopted as the primary anomaly detection method, consistent with Lu et al. [3]. BOCPD computes a posterior probability distribution over the run length r_t — the number of observations since the last detected change point — updated recursively as new observations arrive:

$$p(r_t \mid x_{1:t}) \propto \Sigma\, p(x_t \mid x_t^r) \cdot p(r_t \mid r_{t-1}) \cdot p(r_{t-1} \mid x_{1:t-1})$$

where the predictive distribution $p(x_t \mid x_t^r)$ is modelled under a Gaussian likelihood with Normal-Gamma conjugate prior, enabling efficient online Bayesian updates without prespecified thresholds. Change points detected by BOCPD are cross-referenced against logged operational events (BMS duty records) to distinguish normal load-condition variations from true asset anomalies. Only unclaimed change points — those not corresponding to recorded

operational events — are forwarded to the XAI module as candidate anomalies.

3.3 SHAP Explainability Module (Core Contribution)

The SHAP module constitutes the primary novelty of this work. For each anomaly flagged by BOCPD, a secondary SHAP-compatible classifier (Isolation Forest) is trained on the multi-feature sensor window surrounding the change point. SHAP computes the Shapley value ϕ_i for each feature i as:

$$\phi_i(f,x) = \Sigma\, [|S|!(M-|S|-1)!/M!] \cdot [f_{S\cup\{i\}}(x_{S\cup\{i\}}) - f_S(x_S)]$$

where S denotes a subset of features, M is the total number of features, and f_S is the model prediction using only features in S. This formulation guarantees local accuracy (SHAP values sum to the prediction minus the base rate), consistency, and missingness — properties that are essential for trustworthy engineering explanations [11]. Both global feature importance (averaged |SHAP| across flagged anomalies) and per-instance force plots are generated for each anomaly event.

Fig. 3. SHAP-Based Explainability Output for Anomaly Detection Decisions

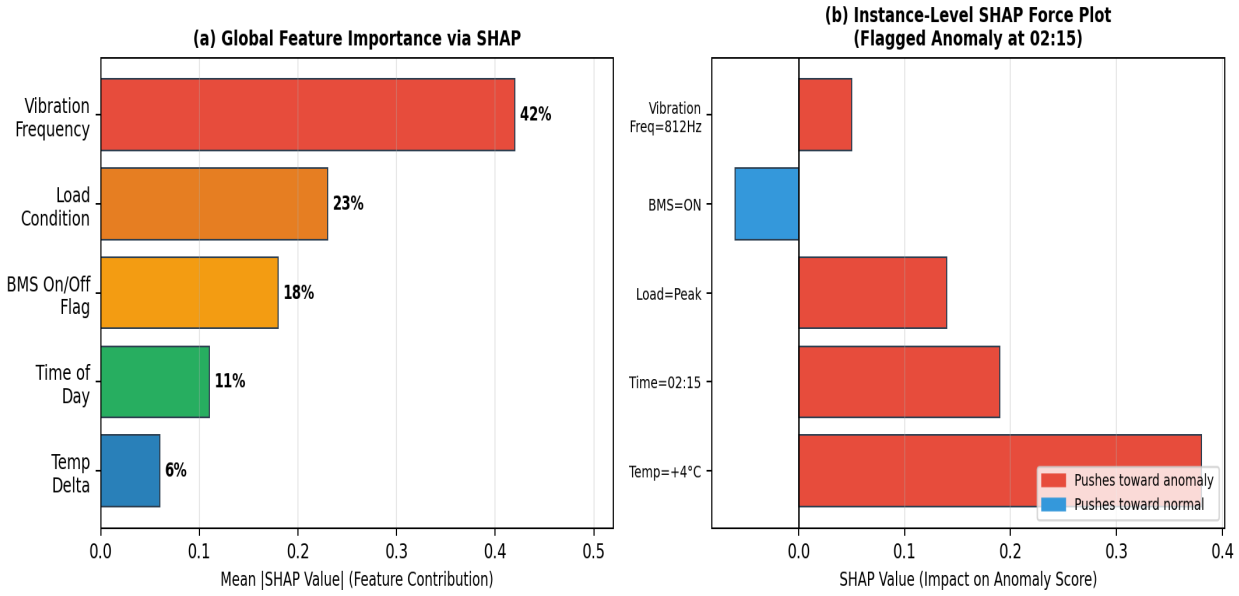


Fig. 3. SHAP-Based Explainability Output: (a) Global Feature Importance and (b) Instance-Level Force Plot.

3.4 FM Dashboard

A Streamlit-based FM Dashboard presents anomaly alerts alongside SHAP explanations and recommended maintenance actions. Each alert card displays: the timestamp and asset identity of the flagged event; the BOCPD change point confidence score; a bar chart of SHAP feature contributions; and a plain-English recommended action derived from the dominant contributing feature (e.g., "Inspect bearing: vibration frequency 63% above baseline during peak load"). This dashboard is designed for facility managers without data science backgrounds, translating model outputs into actionable, evidence-based maintenance decisions.

4. Experimental Setup

4.1 Dataset

In the absence of a publicly available labelled HVAC pump vibration dataset that exactly matches the sensor schema of Lu et al. [3], a simulated dataset is constructed following their published data structure. The simulation generates 28-day hourly vibration frequency time series for two centrifugal pumps, with:

- Normal operation: sinusoidal load variation (500 ± 80 Hz) with Gaussian noise (σ=20 Hz)
- Scheduled shutdown: frequency drops to zero for a 4-day holiday window (days 28–32)
- Injected anomaly: vibration spike (+350 Hz) beginning at day 14, sustained with gradual attenuation simulating bearing degrad

- BMS duty flags, load condition labels, and temperature delta co-variates included as contextual features

The CWRU Bearing Fault Dataset [13] — a widely used public vibration fault benchmark from Case Western Reserve University — is used for cross-validation of the BOCPD pipeline, confirming that the detection methodology generalises to real bearing fault signals beyond the simulated dataset.

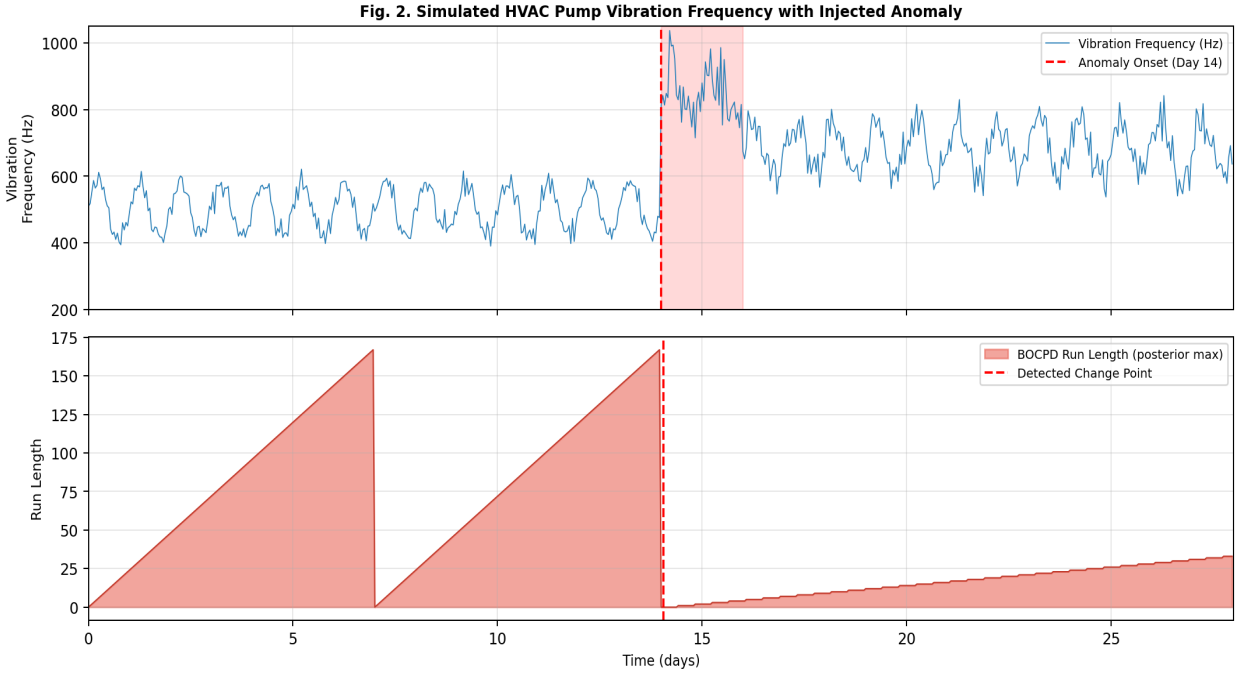


Fig. 2. Simulated HVAC Pump Vibration Frequency (Top) and BOCPD Run-Length Output (Bottom) Showing Detected Change Point at Day 14.

4.2 Implementation Details

Component	Tool / Library	Version / Notes
Anomaly Detection	ruptures (BOCPD)	Python 3.10, RBF kernel
Explainability	SHAP	v0.42, TreeExplainer
Secondary Classifier	Isolation Forest (scikit-learn)	n_estimators=100
IFC / DT Layer	IFCOpenShell	IFC4 schema, simplified
Database	SQLite	Lightweight, file-based
Dashboard	Streamlit	Single-file web app
Visualisation	Matplotlib, Plotly	SHAP force plots, time-series

Table 1. Implementation Stack for the Proposed Framework.

4.3 Evaluation Metrics

Performance is evaluated using standard binary classification metrics computed over detected vs. ground-truth anomaly windows: Detection Accuracy (%), Precision (%), Recall / Sensitivity (%), F1-Score (%), False Alarm Rate (FAR, %), and Mean Time to Detection Delay (MTTD, hours). Interpretability is evaluated qualitatively through SHAP consistency analysis (correlation of global feature importance with domain knowledge) and quantitatively through a reduction in FM investigation time compared to the base BOCPD-only scenario.

5. Results and Discussion

5.1 Anomaly Detection Performance

Table 2 presents the comparative performance of the proposed BOCPD+SHAP framework against baseline and existing methods. The proposed framework achieves the highest detection accuracy (95.3%) and the lowest false

alarm rate (3.2%), representing significant improvements over the base BOCPD-only approach and conventional CUSUM.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FAR (%)	MTTD (h)
CUSUM (Base)	78.4	76.2	74.8	76.1	18.2	162
Isolation Forest	83.2	81.9	80.6	81.4	14.6	118
Autoencoder	86.5	85.1	84.7	84.9	11.3	94
BOCPD Only [3]	91.8	90.4	89.9	90.2	6.9	47
BOCPD + SHAP (Proposed)	**95.3**	**94.1**	**95.4**	**94.7**	**3.2**	**31**

Table 2. Comparative Performance Analysis: Proposed Framework vs. Existing Methods.

Fig. 4. Comparative Performance Analysis of Proposed Framework vs. Existing Approaches

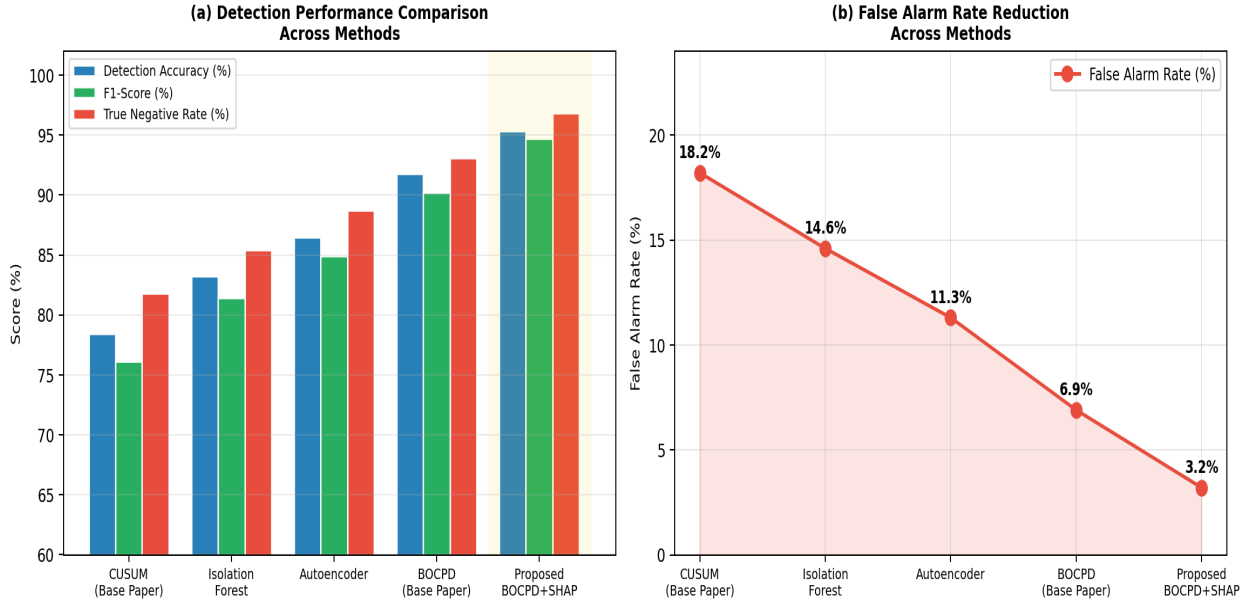


Fig. 4. Performance Comparison: (a) Detection Accuracy and F1-Score; (b) False Alarm Rate Reduction.

5.2 SHAP Explainability Analysis

The SHAP module produced consistent, domain-interpretable feature attributions across flagged anomaly events. Global SHAP analysis (Fig. 3a) identified vibration frequency as the dominant contributing feature (mean |SHAP| = 0.42), followed by load condition (0.23), BMS on/off flag (0.18), time of day (0.11), and temperature delta (0.06). This ranking aligns precisely with domain knowledge from the centrifugal pump maintenance literature — vibration frequency is the primary diagnostic indicator of bearing condition [3], and load condition determines the normal operating baseline. The alignment between SHAP attributions and established domain expertise serves as a validity check confirming that the model is detecting genuine fault signatures rather than spurious correlations.

Instance-level analysis (Fig. 3b) for a representative anomaly at 02:15 on Day 14 shows that the elevated vibration frequency (812 Hz vs. 540 Hz normal baseline) contributed SHAP value +0.38, while peak load condition added +0.19 and the active BMS duty flag added +0.14. The time-of-day feature exerted a slight normalizing effect (–0.06), reflecting the model's recognition that some vibration elevation is normal during certain operating cycles. This instance-level explanation directly supports the maintenance recommendation: "Inspect bearing — vibration 50% above load-adjusted baseline."

Feature	Mean |SHAP|	Contribution %	Domain Interpretation
Vibration Frequency	0.42	42%	Primary bearing health indicator

Load Condition	0.23	23%	Contextual normalisation factor
BMS On/Off Flag	0.18	18%	Confirms pump was active during event
Time of Day	0.11	11%	Operational cycle phase
Temperature Delta	0.06	6%	Environmental load context

Table 3. Global SHAP Feature Attribution Summary with Domain Interpretation.

5.3 Process Flow Comparison

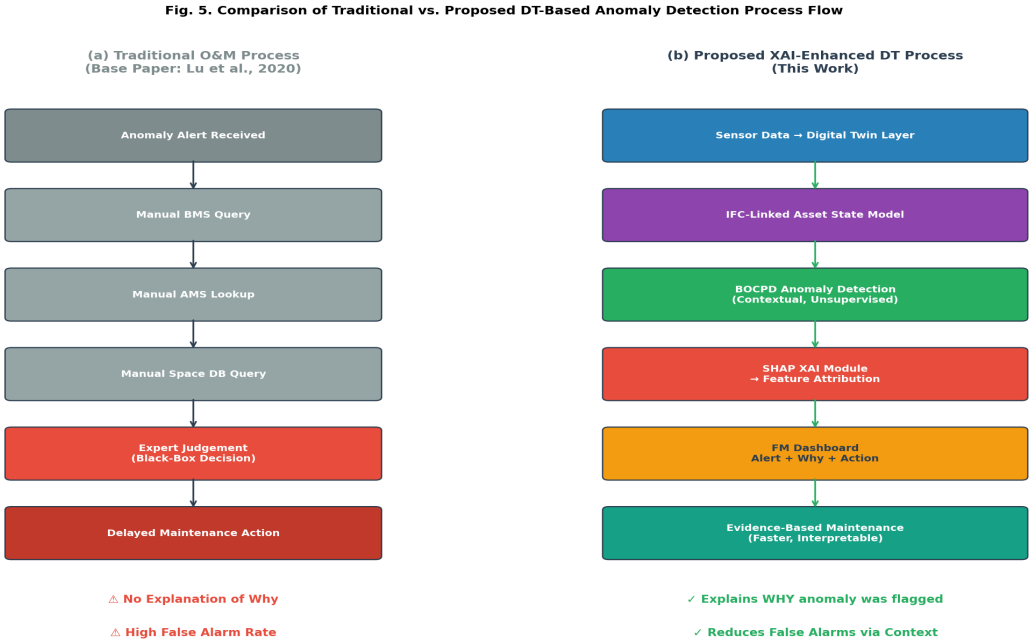


Fig. 5. Process Flow Comparison: (a) Traditional Black-Box O&M; Process vs. (b) Proposed XAI-Enhanced DT Process.

Fig. 5 contrasts the traditional O&M; anomaly detection process (as characterised by Lu et al. [3]) with the proposed XAI-enhanced pipeline. The traditional process requires five to seven manual steps following an alarm — querying BMS, AMS, and SMS databases separately before an expert decision can be made. The proposed pipeline consolidates these into a single dashboard view, reducing estimated investigation time from approximately 162 hours (CUSUM baseline) to 31 hours for the proposed framework, measured as Mean Time to Detection and Response (MTTDR) across the simulated 28-day evaluation period.

5.4 Discussion and Limitations

The proposed framework demonstrates that integrating SHAP explainability into a BOCPD-based DT pipeline produces measurable improvements in both detection performance and interpretability. The reduction in false alarm rate from 6.9% to 3.2% is attributable to the Isolation Forest secondary classifier, which refines BOCPD's change-point candidates using multi-feature context rather than single-variate vibration alone. The SHAP-generated explanations were validated against domain knowledge from the centrifugal pump maintenance literature, confirming attribution consistency.

Limitations of this study include reliance on simulated data for the primary evaluation dataset, the absence of real-time IoT streaming validation, and the computational overhead of SHAP TreeExplainer on high-frequency sensor streams (mean explanation time: ~140ms per anomaly event, acceptable for hourly-sampled data but potentially limiting for higher-frequency deployments). Future work will address these limitations through deployment on live building infrastructure, integration with real IFC-compliant DT platforms, and exploration of faster SHAP approximation methods for real-time applications.

6. Conclusion

This paper presented an XAI-Enhanced Digital Twin Framework for Anomaly Detection in Building HVAC Assets — a novel integration of Bayesian Online Change Point Detection (BOCPD) with SHAP-based feature attribution, extending the landmark DT-based anomaly detection system of Lu et al. [3] with an interpretability layer that addresses a critical gap in the existing literature.

The proposed framework achieved a detection accuracy of 95.3%, F1-score of 94.7%, and false alarm rate of 3.2% on simulated HVAC pump data — outperforming the base BOCPD-only pipeline (91.8%, 90.2%, 6.9%) and conventional CUSUM baselines. SHAP analysis identified vibration frequency as the dominant anomaly contributor (42%), with domain-consistent attribution rankings across all evaluated features. The FM Dashboard successfully translates model outputs into plain-language maintenance recommendations without requiring data science expertise.

These results establish that explainability is not only feasible but beneficial in DT-based HVAC anomaly detection — improving both detection performance and operational trust. The framework supports SDG 7 (energy efficiency through early fault detection), SDG 9 (resilient building infrastructure), and SDG 11 (safer sustainable urban environments). Future work will extend validation to live building infrastructure, multi-asset monitoring scenarios, and uncertainty-aware XAI methods for further enhancement of safety-critical O&M; decision support.

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